Technologies Report — Chikku Robotics

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1. Artificial Intelligence & Machine Learning

> Our AI research and engineering team develops state-of-the-art models for perception, planning, and human-robot interaction. We combine classical robotics with modern deep learning to create truly intelligent machines.

1.1 Deep Learning

Attribute	Detail
Description	Convolutional neural networks (CNNs), transformers, and recurrent architectures optimized for real-time robotic inference on edge devices.

Applications:

1. Object detection and classification
2. Semantic scene understanding
3. Human pose estimation
4. Gesture recognition

1.2 Reinforcement Learning

Attribute	Detail
Description	Model-based and model-free RL for robot control, navigation, and manipulation. Training in simulation with sim-to-real transfer.

Applications:

1. Adaptive navigation in dynamic environments
2. Robotic grasping and manipulation
3. Multi-robot coordination
4. Energy-optimal path planning

1.3 Natural Language Processing

Attribute	Detail
Description	Transformer-based language models fine-tuned for robotic command understanding and multi-turn dialogue in **30+ languages**.

Applications:

1. Voice command interpretation
2. Intent classification
3. Conversational AI for service robots
4. Multilingual translation

1.4 Generative AI

Attribute	Detail
Description	Large language models and diffusion models for task planning, code generation, and robot behavior synthesis.

Applications:

1. Automated task planning from natural language
2. Robot program generation
3. Synthetic training data generation
4. Predictive world modeling

2. Computer Vision

> Our vision systems enable robots to see, understand, and interact with their environment in real time. From 3D perception to facial recognition, our CV stack runs efficiently on embedded hardware.

2.1 3D Perception

Attribute	Detail
Description	Stereo vision, structured light, and LiDAR-camera fusion for accurate 3D environment mapping and object localization.

Technologies:

1. Stereo depth estimation
2. Point cloud processing (PCL)
3. LiDAR-camera calibration and fusion
4. Visual SLAM (ORB-SLAM3)

2.2 Object Detection & Tracking

Attribute	Detail
Description	Real-time multi-object detection and tracking using YOLOv8 and transformer-based detectors, optimized for edge deployment.

Technologies:

1. YOLOv8 (Nano to XL)
2. DeepSORT tracking
3. ByteTrack
4. TensorRT optimization

2.3 Facial & Gesture Recognition

Attribute	Detail
Description	Privacy-preserving facial recognition for personalized robot interactions. Hand and body gesture recognition for intuitive human-robot communication.

Technologies:

1. ArcFace embeddings
2. MediaPipe holistic
3. **Privacy-first:** on-device only, no cloud upload

2.4 Industrial Inspection

Attribute	Detail
Description	Automated visual inspection for manufacturing quality control — defect detection, dimensional measurement, and surface analysis.

Technologies:

1. Anomaly detection (PaDiM, PatchCore)
2. OCR for part numbers
3. Thermal imaging analysis
4. High-res multi-camera calibration

3. Robotics Engineering

> Core robotics capabilities spanning kinematics, dynamics, control theory, and mechatronics design.

3.1 Motion Planning & Control

Attribute	Detail
Description	Advanced trajectory planning and real-time control for manipulators and mobile platforms.

Technologies:

1. Model Predictive Control (MPC)
2. Impedance control
3. Admittance control
4. Whole-body control
5. Trajectory optimization (TrajOpt)

3.2 SLAM & Localization

Attribute	Detail
Description	Simultaneous Localization and Mapping for autonomous navigation in unknown and dynamic environments.

Technologies:

1. LiDAR SLAM (Cartographer, FAST-LIO2)
2. Visual SLAM (ORB-SLAM3)
3. Visual-inertial odometry
4. Multi-sensor fusion (EKF, factor graphs)

3.3 Manipulation & Grasping

Attribute	Detail
Description	Robotic grasping and manipulation for diverse objects in unstructured environments.

Technologies:

1. 6-DoF grasp planning (GPD, GraspNet)
2. Dexterous manipulation
3. Force-feedback assembly
4. Soft robotic grippers

3.4 Mechatronics & Hardware

Attribute	Detail
Description	Custom actuator design, sensor integration, and embedded electronics for our robot platforms.

Technologies:

1. Custom BLDC motor drivers
2. FPGA-based real-time control
3. CAN bus / EtherCAT communication
4. Battery management systems (BMS)

4. Edge AI & Computing

> All our AI inference runs on-device for low latency, privacy, and offline capability. We design custom edge computing hardware optimized for robotic workloads.

4.1 NVIDIA Jetson Orin Integration

Attribute	Detail
Role	Primary edge AI platform
Description	Custom-optimized models run on **Jetson Orin AGX** (275 TOPS) and **Orin NX** (100 TOPS).

4.2 Chikku Edge AI Accelerator

Attribute	Detail
Role	Custom FPGA-based accelerator
Description	Ultra-low-latency perception tasks (**< 5ms inference**). Optimized for safety-critical real-time applications.

4.3 On-Device LLM

Attribute	Detail
Role	Local language models
Description	Quantized language models (4-bit, 8-bit) running locally on robot hardware. Enables private, offline conversational AI without cloud dependency.

5. Cloud & Data Platform

> Our cloud infrastructure supports fleet management, continuous learning, and data analytics across thousands of deployed robots.

5.1 Fleet Management

Monitor, control, and optimize robot fleets at scale. Real-time telemetry, task assignment, and traffic management.

5.2 Digital Twin

Physics-accurate simulation of robots and environments. Test new behaviors, validate safety, and train AI models in simulation before deployment.

5.3 Continuous Learning

Federated learning pipeline that improves robot AI models using field data while preserving privacy. Models improve with every deployment.

5.4 Analytics & Insights

Operational analytics dashboard showing robot utilization, efficiency metrics, anomaly detection, and predictive maintenance alerts.

6. Safety & Certification

> Safety is the foundation of all our technology. We implement multi-layered safety architectures and comply with international robotics safety standards.

6.1 Safety Standards

No.	Standard	Description
1	**ISO 13482**	Safety requirements for personal care robots
2	**ISO 10218**	Industrial robot safety
3	**ISO 13849**	Safety-related parts of control systems (up to PL e / SIL 3)
4	**IEC 61508**	Functional safety of electrical/electronic systems
5	**IEC 60601**	Medical electrical equipment (for MediBot)
6	**CE Marking** (EU), **FCC** (USA), **MIC** (Japan)	Regional certifications

6.2 Safety Features

No.	Feature
1	Multi-layer safety: hardware safety controller + software safety monitor
2	Force-limited joints with real-time collision detection
3	360° LiDAR-based safety zones with configurable protective fields
4	Emergency stop with **< 50ms response time**
5	Fail-safe braking on all mobile platforms
6	Redundant light detection and ranging for outdoor vehicles